

AI solutions deployment in Azure

The strategy you choose depends on whether you're building a quick prototype, an internal tool, or a high-traffic product.

1. The "Community" Way (Fast & Free)

If you just want to share your app with a few people or build a portfolio. E.g. Streamlit community

2. The "Serverless" Way (Scale to Zero)

For modern AI apps, **Container-as-a-Service (CaaS)** is the gold standard. It's cost-effective because the app "sleeps" when no one is using it and wakes up instantly when a request comes in.

Cloud	Service Name	Why use it?
Azure	Container Apps	Great for LangChain apps; easy secret management.
GCP	Cloud Run	Generally considered the easiest to use; excellent scaling.
AWS	App Runner	Simplifies the complex AWS ecosystem for web apps.

What is a Container?

In software, a Container is a lightweight, standalone package that includes everything your app needs to run:

- The code (app.py).
- The language (Python 3.11).
- The libraries (Streamlit, LangChain).
- The system settings.

When you put your AI app in a container, it will run exactly the same on your laptop, on Azure, or on a private server.

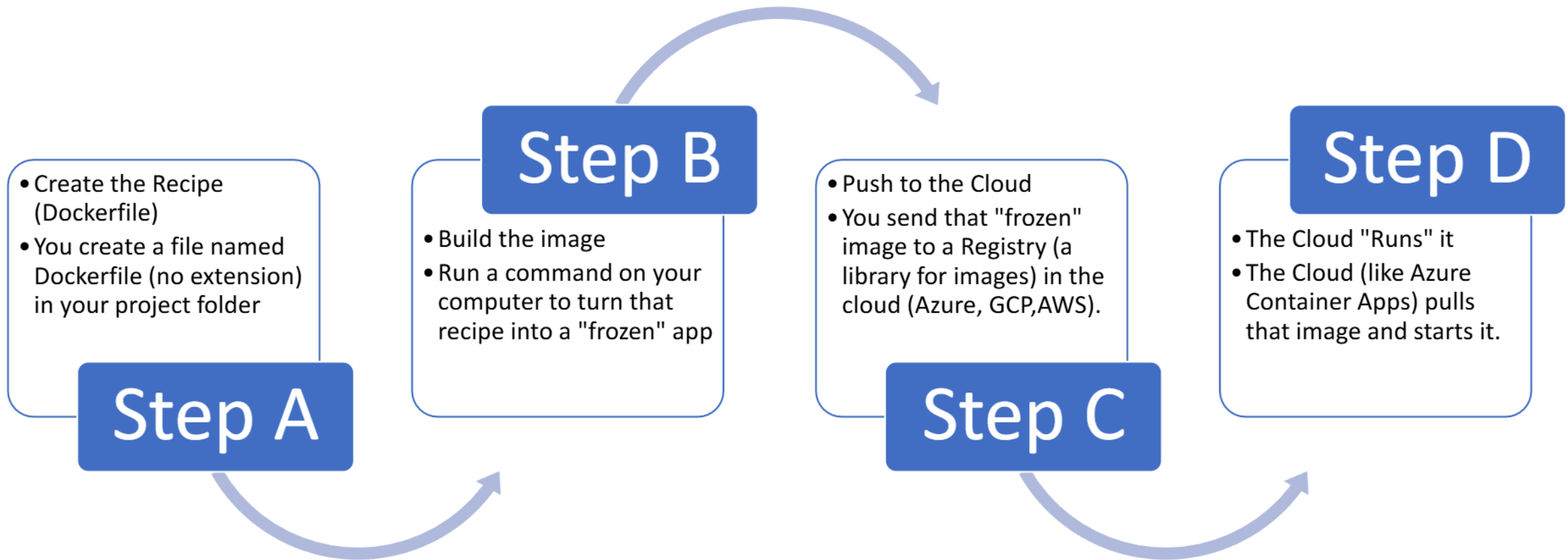
Think of a **shipping container** on a boat. It doesn't matter if the cargo inside is furniture, electronics, or food; the container is a standard size, and every crane in the world knows how to move it.

What is Docker?

If a Container is the "box," then Docker is the "tool" used to build, seal, and ship that box.

- Dockerfile: This is a simple text file (a recipe) that tells Docker how to build the container.
- Image: When you run the Dockerfile, it creates an "Image." Think of this as a "frozen" version of your app.
- Container: When you actually "run" that Image, it becomes a live Container.

How the Workflow actually looks



Advantage – It is “Serverless”

When you use a service like Azure Container Apps or Google Cloud Run, the cloud is smart.

- Idle: If no one visits your website, the cloud "kills" the container. You pay \$0.
- Request: When someone types your URL, the cloud quickly starts your container (the "wake up").
- Scaling: If 1,000 people visit at once, the cloud just makes 10 copies of your container instantly.

Summary

Terms:

- Docker: The tool used to create the package.
- Image: The "executable" version of your app (the file).
- Container: The "running" instance of your app (the process).
- Registry: A storage unit in the cloud for your Images.

Strategy:

- If it's for your portfolio: Use Streamlit Community Cloud.
- If it's for a client or business: Use Azure Container Apps or Google Cloud Run.